Supplementary Materials

Transfer learning with simulated variants and calculated quantities for nanobody melting-temperature prediction

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Supplementary methods

Experimental target data and target-label scaling

All target-task comparisons use the same NbBench thermo-seq endpoint [1]. The processed tables hold an amino-acid sequence field and a melting-temperature label in degrees Celsius. The split sizes are 57 training, 114 validation, and 396 held-out test sequences. As described in Methods, we deliberately assigned the smallest NbBench partition to training and the largest to the test set so that the model sits in the low-data regime the study targets.

Split definitions. The splits are fixed throughout. The 57 training examples fit network parameters, the validation split drives checkpoint selection, early stopping, and choice among candidate settings, and the test split stays untouched until those choices are fixed. The same rule applies to Tm-only models and to models trained with computational labels.

Target-label scaling. Before optimization, the training workflow maps the experimental Tm values to a scaled regression target. The scaler is fit only on the 57 training labels, using robust scaling followed by linear rescaling to [0,1]. Validation labels are transformed with this training-fitted scaler during model selection. Test predictions are averaged across seeds in scaled-label space and inverse-transformed to degrees Celsius before errors are computed, so the validation and test label distributions never influence the fitted transform.

Experimental-label scaling reference. This reference uses the same training split but subsamples it to the requested number of labels with the run seed; the validation and test splits are never subsampled. It serves only as a positive control for the pipeline. Adding true target labels should lower the Tm prediction error.

Processed computational-label tables

Computational labels are loaded as computational tasks rather than appended to the Tm data as extra measurements. Table 1 lists the processed target and computational-label files used in the reported comparisons. Mutation-effect tables carry sequence fields and [0,1]-scaled computational-label fields. The MD native-contact tables use the same processed format so that they can enter the same computational-task loader; there, the scaled label is a [0,1]-scaled native-contact fraction rather than a mutation free energy. The FEP-matched MD table (the primary MD label) is built on the same 1MEL/4IDL mutation scans as FEP; the diverse-panel MD table provides the heterogeneous simulation-acquisition comparison.

Table 1 Processed target and computational-label data. Row counts exclude header rows.

Data type	Role	Processed rows	Use in reported comparisons
Experimental Tm	Target train	57	Model fitting
Experimental Tm	Target validation	114	Checkpoint and setting selection
Experimental Tm	Target test	396	Final held-out evaluation
FEP mutation free energy	Computational tasks	435 + 409	Main computational task
MD native-contact (FEP-matched scan)	Computational tasks	431 + 406	Matched-design MD label (primary)
Rosetta mutation score	Computational tasks	435 + 409	Computational-label comparison
ThermoMPNN stability score	Computational tasks	435 + 409	Computational-label comparison
Random variants scored by Rosetta	Computational tasks	1000 + 1000	Computational-label comparison
ESM2-proposed variants scored by Rosetta	Computational tasks	1000 + 1000	Computational-label comparison
MD native-contact (diverse panel)	Computational task	1143	Heterogeneous simulation-design comparison

Computational-task assignment. The loader assigns each row to a target or computed-label task. Experimental Tm rows use the Tm head. The 1MEL and 4IDL tables use either separate computational heads or one head conditioned on a learned structure indicator, according to the setting selected for that label source. This assignment fixes which regression head and loss term each row uses.

Sequence-overlap check. We checked exact sequence overlap between the processed computational tables and the fixed NbBench splits. The FEP, Rosetta, and ThermoMPNN mutation-effect tables, the FEP-matched MD native-contact table, and the random and ESM2-proposed variant tables — all built on the 1MEL and 4IDL scaffolds — had no exact match to any NbBench training, validation, or test sequence. The diverse-panel SAbDab MD table contained exact matches to 2

NbBench training, 4 validation, and 8 test sequences. We disclose this because it bears on the simulation-design comparison. The diverse panel gives a small frozen-encoder gain, so the overlap is a limitation of that secondary effect and is one reason not to interpret the matched-versus-heterogeneous contrast as a fully isolated causal experiment. It cannot explain the matched-scan result, whose sequences have no exact NbBench overlap.

Computational-label sampling. For mutation-effect experiments with requested count n, the loader samples up to n rows independently from each of the two template tables using the run seed; the final FEP setting with $n = 320$ therefore supplies 320 sampled 1MEL rows and 320 sampled 4IDL rows before the computational-task split. The FEP-matched MD native-contact label samples from the same two template-specific tables. The diverse-panel MD control samples n rows from its single table. Computational-label rows are split 80/20 into computational-training and computational-validation rows with the same seed. The production training call then filters the validation data passed to the trainer so that checkpoint selection and early stopping depend only on the experimental Tm validation examples.

FEP-table provenance and charge-change check

The retained FEP tables contain Ala, Asp, Ile, and Gln mutation scans. The 1MEL table contains 116, 80, 119, and 120 rows from these scans, respectively; the 4IDL table contains 105, 84, 113, and 107 rows. Row-level provenance was checked against the archived scan outputs. The available Phe scans were not included in the retained tables.

We also estimated the analytical periodic net-charge correction for charge-changing mutations. Of the 844 retained rows, 284 change the formal charge. For a cubic 70-Å box, the correction magnitude used in this check was $0.086(\Delta q)^2$ kcal mol^{-1} at a solvent dielectric of 78. We applied the correction before repeating the same sign reversal, robust scaling, Yeo–Johnson transform, and min–max scaling used for the model-facing labels. The corrected and uncorrected labels had a correlation of 0.99989; the largest scaled-label shift was 0.008, and no row shifted by more than 0.02. Larger per-charge sensitivity values are shown only as bounds in Supplementary Fig. S4; they are not estimates of the omitted correction terms.

Diverse-panel MD control: SAbDab structure selection

The following describes the *diverse-panel* MD native-contact comparison used for the simulation-design contrast in Fig. 2. The FEP-matched MD label used for the main result is built on the 1MEL/4IDL mutation scans as described in Methods. The diverse-panel MD control was built from SAbDab, the Structural Antibody Database [2]. The archived SAbDab nanobody summary table holds 2422 rows with PDB identifier, heavy- and light-chain assignments, model index, antigen annotation, experimental method, resolution, species, and other fields. The manifest workflow matches this summary against the archived IMGT-renumbered PDB files.

Structure-selection rule. The rule is deterministic. For each PDB identifier with both a SAbDab summary row and an IMGT-renumbered PDB file, the workflow takes the first heavy-chain row in the SAbDab table and accepts the chain only if that chain identifier is present in the IMGT PDB file. The output manifest holds one entry per PDB identifier, with fields for the manifest row number, PDB identifier, selected heavy chain, SAbDab TSV row, IMGT PDB path, and MD run directory. This initial one-per-PDB manifest contained 1177 structures.

400 K simulation panel. The 400 K simulations reused the systems prepared for the 300 K runs and built a 1146-entry 400 K manifest, consisting of 798 X-ray, 345 electron-microscopy, and 3 NMR structures according to the SAbDab method field. None of the selected rows had an annotated light chain, consistent with a single-domain antibody panel. After trajectory and Q-value processing, 1143 unique PDB identifiers yielded usable computational-label rows.

MD system preparation and simulation protocol

System preparation. Each structure was prepared as a nanobody-only system. The workflow takes the IMGT-renumbered PDB as the input structure, extracts the selected heavy chain, keeps only protein atoms, and leaves the termini uncapped, at pH 7.4. Chains are cleaned with PDBFixer, and pH-aware protonation is assigned through pdb2pqr/PROPKA when available [3, 4], with a fallback to AmberTools-based hydrogen assignment and Amber naming. The prepared chain is solvated in explicit OPC water with 15 Å padding and 0.15 M NaCl, and Amber topology and coordinate files are built with ff19SB and OPC [5–7].

300 K equilibration branch. The base 300 K branch uses OpenMM on CUDA devices [8]. It runs 1 ns of NVT followed by 1 ns of NPT at 300 K and 1 atm, with a 2-fs timestep and no hydrogen-mass repartitioning. Production at 300 K was generated for the broader study, but the computational label used here comes from the 400 K branch.

400 K production branch. The first 400 K relaxation stage restarts from the 300 K equilibrated state and runs 1 ns of NVT at 400 K with Cα restraints; the second repeats 1 ns of restrained 400 K NVT. Both relaxation stages use a 2-fs timestep and no hydrogen-mass repartitioning. Production runs 100 ns of restraint-free 400 K NVT, with hydrogen-mass repartitioning, 4 amu hydrogen mass, a 4-fs timestep, and coordinate/energy output every 100 ps.

Integrator and reporting settings. In OpenMM, explicit-solvent periodic systems use PME electrostatics with a 1.0-nm nonbonded cutoff and constraints on bonds to hydrogen. NPT stages use a Monte Carlo barostat; NVT stages omit it. The integrator is the Langevin middle integrator with 1 ps^{-1} friction. The 100-ps reporting interval gives 1000 expected frames for a complete 100 ns production trajectory.

MD-derived Q-value extraction

Trajectory input requirements. The processed MD table was generated from the archived 400 K production trajectories. A system was eligible only if it had a production trajectory, a prepared reference structure, and an Amber parameter file. The amino-acid sequence was read from the prepared structure using the first Cα atom of each residue, and sequences shorter than 50 residues were excluded.

Native-contact definition. Q-values are computed with MDTraj [9]. The topology is loaded from the Amber parameter file, and the native reference is frame 0 of production. The calculation uses protein backbone heavy atoms. A candidate native contact must span more than three residues, have a reference-frame distance below 0.45 nm, and include at least one atom from a hydrophilic residue (Asp, Glu, Gln, Asn, Arg, Lys, or His, including Amber protonation variants). For a native-contact pair (i, j) with reference distance r_{ij}^0 and instantaneous distance r_{ij}, the per-frame contribution is

$$q_{ij} = \frac{1}{1 + \exp\{\beta(r_{ij} - \lambda r_{ij}^0)\}},$$

with $\beta = 50$ nm^{-1} and $\lambda = 1.8$, following the smooth native-contact form of Best-Hummer-style Q-value analyses [10]. The frame-level Q-value is the mean of q_{ij} over all selected native contacts.

Trajectory-level Q-value label. The trajectory-level label is the mean Q-value over the terminal portion of production. For trajectories with more than 300 frames, the implementation uses the larger of 300 frames and one third of the available frames; for shorter trajectories it uses all frames. For a complete 100 ns trajectory with 100-ps output, this is 333 terminal frames, roughly the final 33 ns. In the processed 400 K table, the number of frames used ranges from 66 to 333 (median 333), the number of native contacts from 28 to 728 (median 173), and the sequence length from 58 to 461 residues (median 122). Raw Q-values range from 0.0437 to 0.9994 and are scaled to [0,1] for computational-task training.

Sequence tokenization and encoder inputs

Sequences are tokenized with the ESM2 tokenizer for the selected encoder [11], using truncation and fixed-length padding to 160 residues. In the production protocol, the trainer receives all target and sampled computational-label rows for training, but its evaluation data are filtered to the experimental Tm validation rows.

Hot-encoder input path. The hot-encoder setting keeps tokenized sequences in the training data and updates the ESM2 weights. The frozen-encoder setting runs ESM2 once per target and computational-label sequence to precompute first-token embeddings, after which the optimizer updates only the regression trunk and task heads.

Neural-network architecture

The default encoder is the 8M-parameter ESM2 model, and the model uses its first-token representation as the sequence-level embedding. The shared trunk has dimensions hidden-size $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 32, with ReLU activations and dropout after the first two hidden layers. The Tm head maps the 32-dimensional representation to one scaled scalar prediction.

Target and computational heads. The candidate models use a Tm path and, for transfer models, one or more computed-label heads. For two-table sources, the search compares separate structure-specific heads with a head conditioned on a learned structure indicator. Every output head returns one scaled scalar.

Architecture search. The selected model form can be shared, residual, or latent. The shared form sends all rows through one 256–128–32 trunk. The residual and latent forms use separate 256–128–32 paths and combine them in the Tm prediction. The final FEP models use the shared form with separate heads. Other sources select different forms in some encoder settings; these selected forms are listed in Supplementary Fig. S2.

Training objective, optimizer, and checkpoint selection

Each training example is a sequence, a scaled label, and a task identifier, and contributes loss only through the regression head for its task. The per-task loss is Huber loss with $\delta = 1.0$ on the scaled label. The default multi-task objective uses learned task-uncertainty weights [12]. For task k with loss L_k and learned log-scale s_k, the contribution is

$$\frac{L_k}{2\exp(2s_k)} + s_k,$$

and the total loss sums the contributions from the tasks present in the minibatch. Fixed computational-loss weights are also implemented and were included in the candidate searches for computational classes where loss balance could shift validation performance.

Optimizer and schedule. The optimizer is AdamW. The default head/trunk learning rate is 3×10^{-4}, the default hot-encoder learning rate is 1×10^{-4}, weight decay is 0.04, dropout is 0.15, batch size is 16, warmup is 100 steps, and the schedule is cosine decay over at most 400 epochs. Checkpoints are evaluated and saved once per epoch, and mixed precision is used when CUDA is available. In the hot-encoder setting, the trainer creates separate optimizer parameter groups for ESM2 parameters and for the freshly initialized trunk/head parameters; in the frozen-encoder setting, ESM2 parameters are non-trainable and no encoder group is used.

Checkpoint selection. Within each run, the best checkpoint has the lowest mean squared error on the experimental Tm validation rows, and early stopping monitors the same metric with patience 15 and threshold 0. Computational-task validation rows are not passed to the trainer in the production protocol. This detail is central to the comparison. Computed labels affect the shared or coupled regression parameters through the training loss, but they do not decide which checkpoint is used for Tm prediction.

Candidate-setting selection and final evaluation

Candidate settings are selected on the experimental Tm validation split, tuning every source independently and separately for the frozen and hot encoder regimes with a two-stage search. Stage 1 fixes learning rate, dropout, and weight decay at defaults and searches the model architecture (shared trunk, residual fusion, or latent fusion) crossed with the auxiliary-head coupling (separate template-specific heads or a template-conditioned shared head). Stage 2 holds the best Stage 1 skeleton and searches, for the hot regime, the encoder learning rate (1×10^{-4}, 5×10^{-5}, 3×10^{-5}, 1×10^{-5}) and, for the frozen regime, the head learning rate (1×10^{-3}, 5×10^{-4}, 2×10^{-4}), each crossed with dropout (0.05, 0.15, 0.30) and weight decay (0.02, 0.10). The completed search records comprise 71 Stage 1 runs and 273 Stage 2 candidates: 115 frozen-encoder and 158 fine-tuned-encoder candidates. Figure S2 shows the completed Stage 2 candidates used for selection rather than the size of the intended Cartesian grid.

Seeded candidate evaluation. Each candidate is evaluated as a three-seed ensemble on the Tm validation split at $n = 320$ labels per scaffold table, and the single best configuration for each source and regime is then evaluated on the held-out test split as a five-seed ensemble. The run seed controls Python, NumPy, PyTorch, and CUDA randomness when CUDA is available, as well as target-training subsampling in the target-label scaling experiment, computational-row sampling, and computational-label train/validation splitting. The label-count curves use the selected per-source configuration while the number of computational labels is varied over 20, 80, 160, and 320.

Statistics and stored outputs

The primary metric is MAE in degrees Celsius on the 396 held-out test examples. For each condition, the five seeded models produce scaled Tm predictions for every test sequence; these are averaged in scaled-label space and inverse-transformed to degrees Celsius. Root mean squared error, coefficient of determination, and Spearman and Pearson correlations are also computed, but the figures emphasize MAE because it reads directly as an average temperature error.

Bootstrap intervals. Single-condition confidence intervals come from nonparametric bootstrap resampling over test examples. The absolute-error vector from the five-seed ensemble is resampled with replacement 10,000 times using a fixed bootstrap seed, and the reported 90% interval is the 5th to 95th percentile range of the bootstrapped MAE values. The 90% interval is used for consistency with the directional paired comparisons in the main text; the stored bootstrap summaries allow other interval levels to be recomputed. Paired comparisons use the same bootstrap indices for the candidate and reference absolute-error vectors. The reported ΔMAE is candidate minus reference, so negative values mean lower error than the reference.

Stored run summaries. Each training and evaluation call writes a structured summary recording the command-line arguments, relevant environment variables, resolved model settings, per-label-count metrics, bootstrap intervals, per-example absolute errors, and paired bootstrap summaries. Compact tables derived from these summaries feed the main computational-label comparison, the frozen-encoder controls, the computational-head controls, and the computational-combination controls.

Interpreting the label-count curves

Target-label scaling reference. The target-label scaling curve is a positive control on the pipeline. It tests whether the same model family and evaluation code recover the expected gain when more true experimental Tm labels are available. The FEP computational-label curve tests something different. It asks whether more labels from a physically related but distinct computed quantity improve the same held-out Tm endpoint, while the FEP-matched MD native-contact curve tests whether a structural stability observable responds to a matched simulation-acquisition design. The decisive comparisons are therefore not "simulation versus no simulation," but whether the simulations cover relevant sequence changes and whether the physical quantity improves models selected with Tm validation data.

Supplementary figure data tables

The supplementary figures are generated from tabular computational-label files derived from the same run summaries as the main figures. The processed computational tables, rendered supplementary figures, and figure-generation code will be distributed with the public analysis archive.

Fig.-to-data and code mapping

The manuscript figures are regenerated from compact result summaries and computational-label tables rather than from scratch run directories. The mapping below is the first place to check when auditing or extending a figure.

Table 2 Manuscript Fig.-to-data mapping. The table identifies the processed summaries and scripts used to regenerate each display item.

Display item	Inputs	Script	Primary comparison
Fig. 1	Conceptual protocol	`plot/make_outline_figures.py`	Multi-task training with Tm-based model selection
Fig. 2	Simulation-plan schematic; design-by-regime paired MD effects; frozen paired label-count curves	`plot/make_outline_figures.py`	Simulation-design axis: matched local scans show a larger frozen native-contact gain
Fig. 3	Source-by-regime paired-effect bars; direct FEP-minus-MD contrast; fine-tuned paired label-count curves	`plot/make_outline_figures.py`	Physical-observable axis: FEP transfers in both regimes, MD native-contact only frozen
Supplementary Figs. S1 to S4	Supplementary manifest and generated TSV tables	`plot/make_supplementary_figures.py`	Data and overlap checks, validation selection, transfer controls, and FEP charge-change checks

Generated supplementary tables. Compact tables record the validation candidates, selected architecture and head, held-out errors, paired comparisons, data-set diagnostics, and FEP charge-change checks. A panel-level mapping table for all supplementary figures is included in the public analysis archive.

Supplementary figures

References

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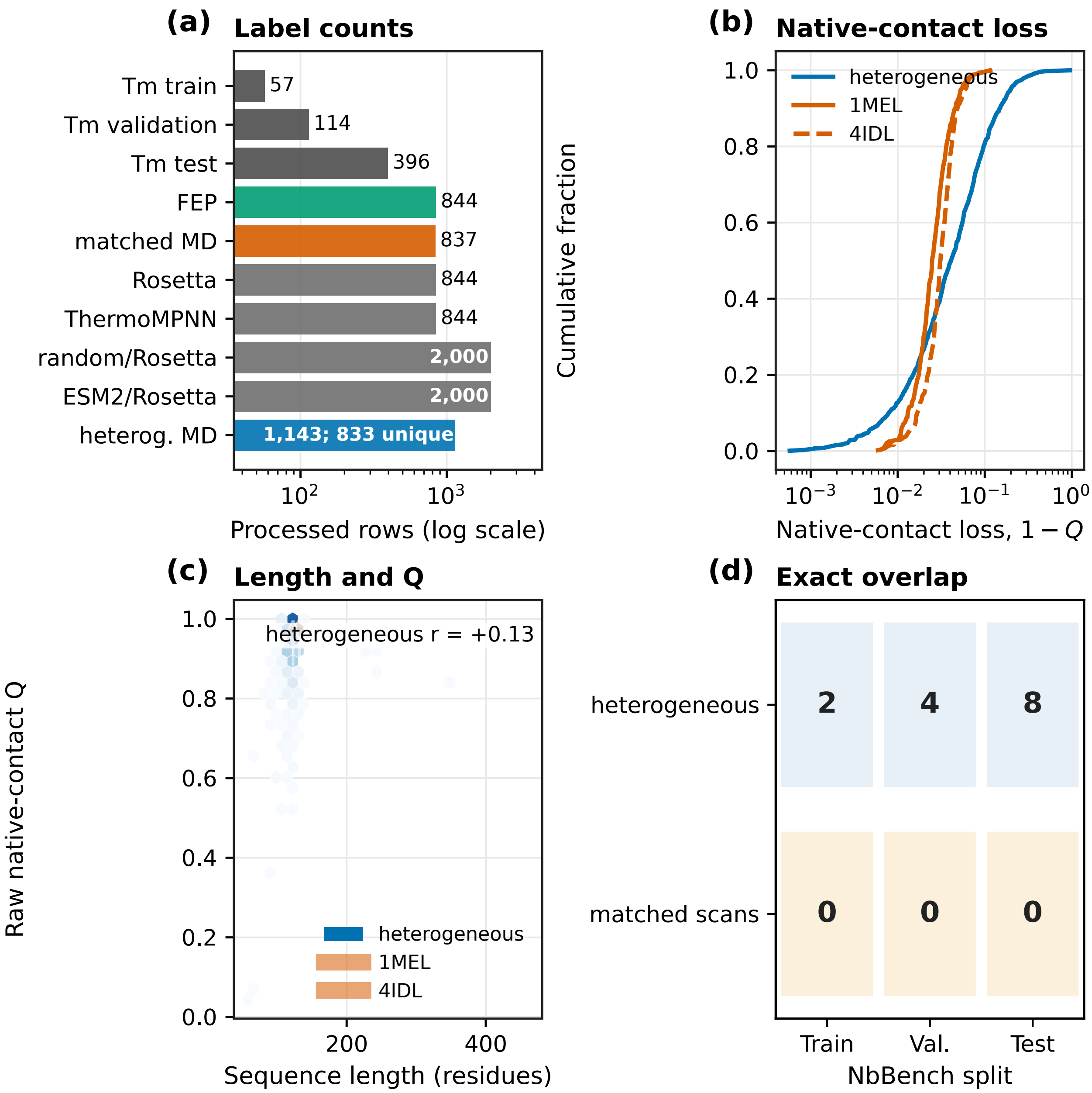

Figure S1 Data sets and diagnostics for the two MD acquisition plans. **(a)** Processed row counts. The heterogeneous MD panel contains 1,143 structure rows but 833 unique sequences; the two scaffold-table counts are recorded in the archived figure data table. **(b)** Empirical cumulative distributions of native-contact loss, $1 - Q$, on a log axis. The extraction details are not identical, so the distributions are descriptive. **(c)** Raw Q versus sequence length. Hexagons show density in the heterogeneous panel; violins show the two fixed-length matched scans. The heterogeneous panel spans 58–461 residues and has Pearson $r = +0.13$. **(d)** Unique exact sequences shared with the three NbBench splits. The matched scans have no matches; the heterogeneous panel has 2, 4, and 8 unique matches.

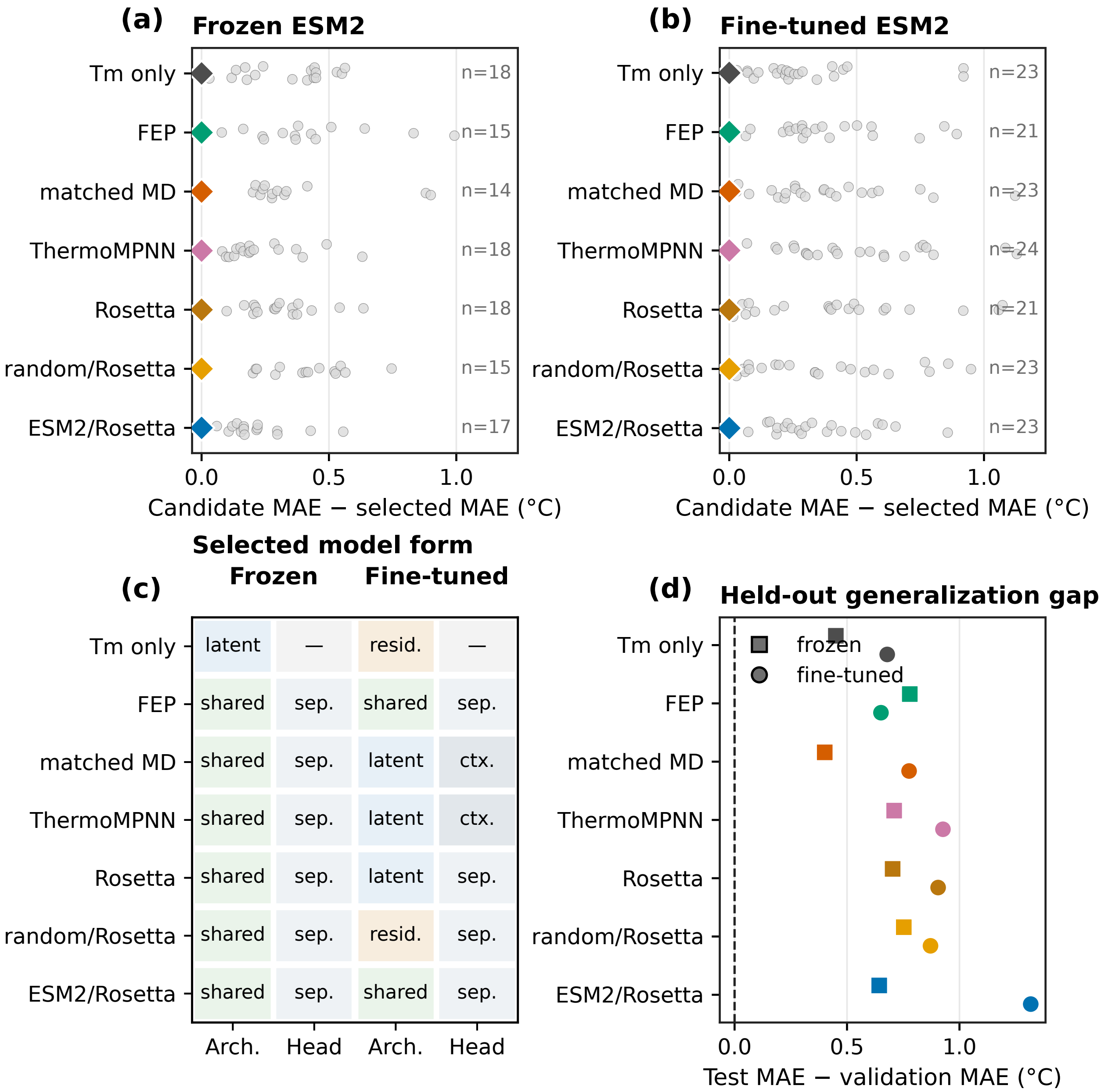

Figure S2 Per-source model selection using experimental Tm validation data. **(a,b)** Completed Stage-2 candidates for frozen and fine-tuned ESM2, shown as validation MAE above the selected candidate for that source. Each small point is one candidate, diamonds at zero mark the selected setting, and row-end labels give candidate counts. The two panels contain 115 and 158 candidates, respectively; no test errors enter selection. **(c)** Architecture and computational-head form selected for every source and encoder setting. A dash denotes a Tm-only model without a computational head. **(d)** Held-out generalization gap, test MAE minus selected validation MAE. Squares and circles denote frozen and fine-tuned ESM2.

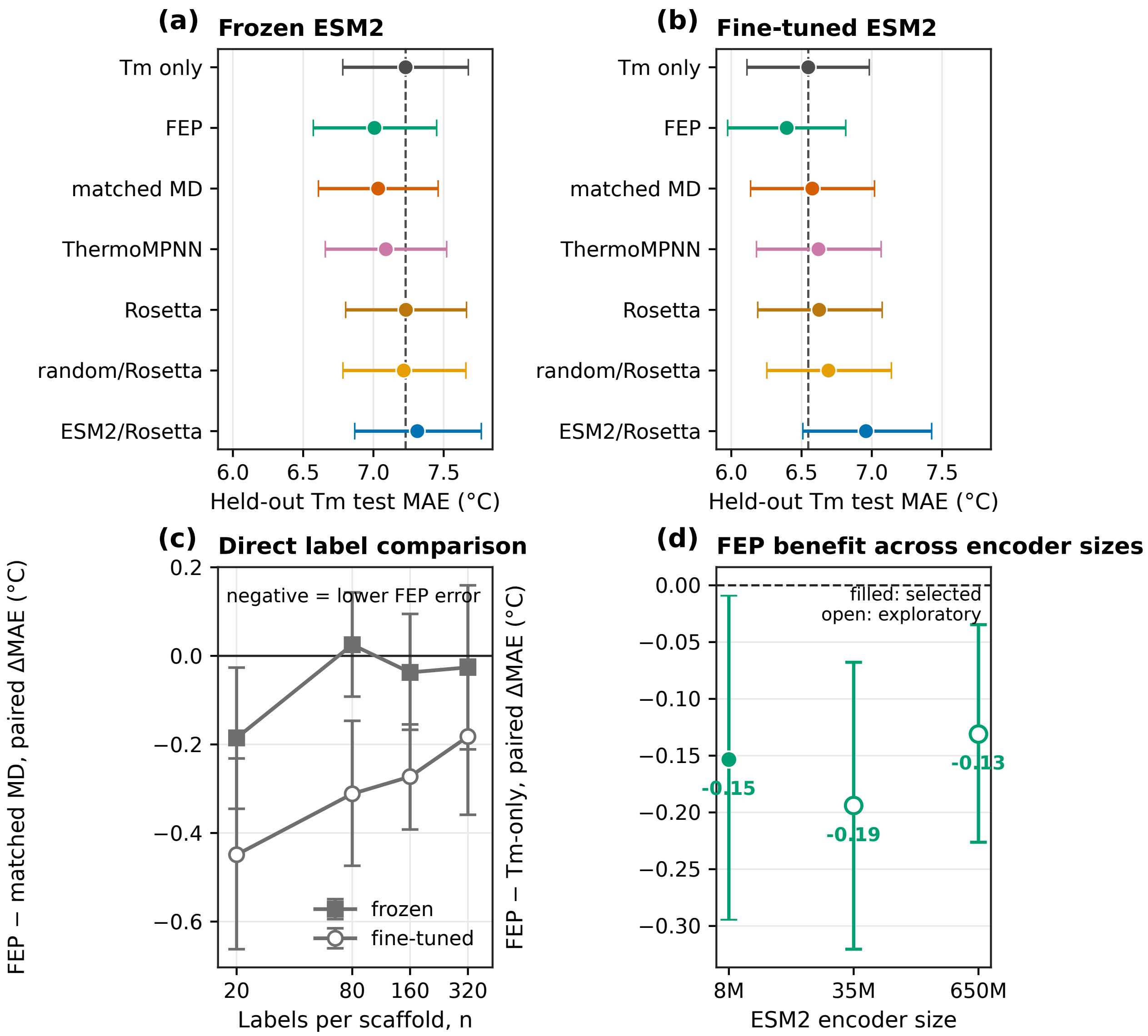

Figure S3 Absolute test errors and transfer controls. **(a,b)** Held-out Tm test MAE for every selected source with frozen and fine-tuned ESM2. Whiskers are 90% bootstrap intervals over the 396 test proteins. **(c)** Direct paired difference between FEP and matched-MD absolute errors at each label count on a $\log_2$ count axis. Negative values favor FEP; whiskers are nominal 90% paired bootstrap intervals. **(d)** Paired FEP-minus-Tm-only MAE for fine-tuned ESM2 sizes. FEP reduces MAE within all three size controls. The 8M point is the staged-selected five-seed model; 35M and 650M are exploratory fixed-configuration ten-seed controls and were not subjected to the same search. Neither larger Tm-only control reaches the absolute MAE of 8M+FEP.

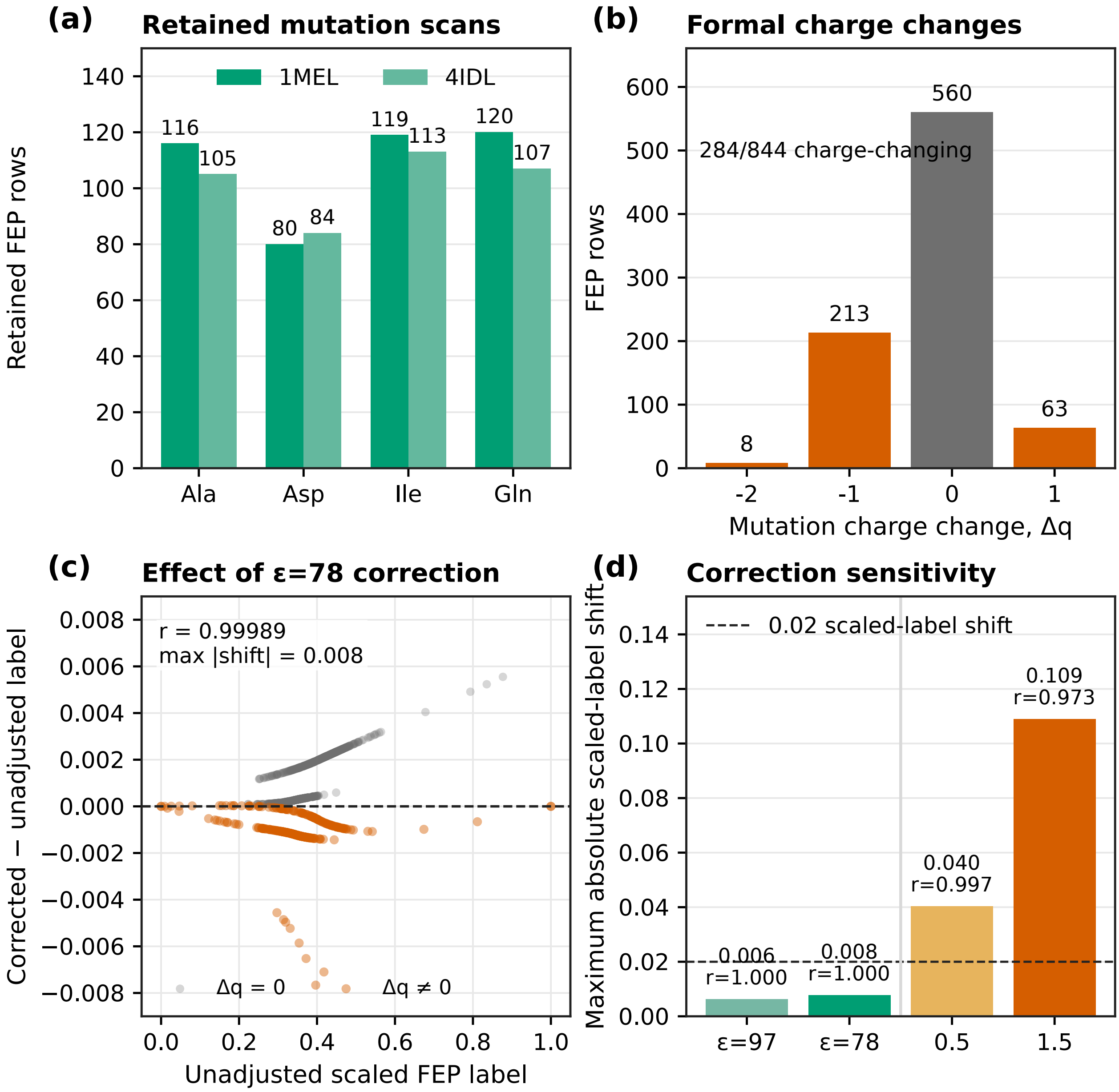

Figure S4 Provenance and charge-change checks for the retained FEP labels. **(a)** Rows from the retained Ala, Asp, Ile, and Gln scans, grouped by structure. **(b)** Distribution of formal mutation charge changes; 284 of 844 rows change charge. **(c)** Per-label shift after the analytical periodic net-charge correction at dielectric 78, plotted against the unadjusted scaled label. The dashed line marks zero shift. **(d)** Maximum scaled-label shift under two periodicity estimates and two larger sensitivity values in kcal mol^{-1} per $(\Delta q)^2$. Numbers above the bars give the maximum shift and corrected-versus-uncorrected correlation. The larger values are sensitivity bounds, not estimates of omitted correction terms.